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Descriptive Analytics
(BCADSN23201)

ASSIGNMENT

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SUBJECT: DESCRIPTIVE ANALYTICS

LAY'S SALES & SNACK DEMAND ANALYTICS

Descriptive Analytics Report

Report Focus Keywords:

Top-Selling Lay's Flavors, Revenue Concentration, Unit Volume Trends, Regional Demand Distribution, Customer Payment Preferences, Pack Size Allocation, Packaging Type Share, Average Order Value (AOV), Descriptive-to-Predictive Transition.

100 Orders · 2,850 Cases Dataset: Lay's Sales Transaction Data	**01 Jan 2025 – 01 May 2025** Time Period	**$83,230 USD** Gross Revenue	**BCA Data Science / AI** Academic Context

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The report follows a structured, enterprise-grade descriptive analytics framework:-

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Note on Sales Data

This descriptive sales report is computed strictly from the verified Lay's Sales Transaction Dataset, representing exactly 100 unique wholesale purchase orders placed between January 1, 2025, and May 1, 2025. Over this analytical timeframe, 2,850 physical cases were purchased, generating a gross recorded turnover of $83,230 USD.

Each row in the dataset represents a completed commercial transaction containing Order_ID, Customer_Name, Country, Lays_Flavor, Pack_Size, Packaging_Type, Quantity purchased (cases), unit Price in USD, Sale_Date, Payment_Method, and calculated Revenue. All metrics reported in this document are direct aggregations and statistical summaries of these verified fields.

Sales Data Completeness & Quality Audit

The transaction logs contain zero missing cells, zero duplicate transactions, and strict data type integrity. Primary keys span Order_ID 1 to 100. The dataset provides an ideal baseline for evaluating commercial snack-hardware performance and wholesale purchasing behavior.

Research Overview & Analytical Framework

Descriptive analytics summarizes historical sales transactions to answer "what sold?", "who purchased?", and "how much revenue was generated?" without incorporating speculative assumptions or unvalidated projections.

Analytical Dimension	Operational Sales Dataset Implementation	Commercial Interpretation
Product Revenue Performance	Grouping gross sales by 6 distinct Lay's flavors	Identifies highest gross revenue flavor contributors.
Physical Unit Velocity	Summing physical case quantities per SKU & month	Measures inventory velocity and consumer adoption.
Transaction Value Spread	5 binned order-value tiers ($300 intervals)	Quantifies basket sizes from single retailer to bulk distributor.
Pack Customization	Pack size & packaging type aggregations	Measures consumer preferences for format and portioning.
Geographic Channel Reach	Cross-border country aggregations	Tracks geographic demand concentration across global territories.

Objective & Core Business Questions

The objective of this descriptive sales analysis is to extract empirical snack-hardware insights and commercial purchasing patterns across flavors, channels, and geographies.

- **Question 1 (Top-Selling Flavors):** Which Lay's flavors generate the highest dollar turnover versus the highest physical case volume? (Section 1)
- **Question 2 (Unit Volume Trends):** How do physical case demand and revenue fluctuate from January to May 2025? (Section 2)
- **Question 3 (Average Order Value):** What is the distribution and spread of order transaction values across buyer profiles? (Section 3)
- **Question 4 (Pack Size Allocation):** Which pack size delivers the optimal balance of unit volume and gross revenue? (Section 4)
- **Question 5 (Packaging Type Share):** How do packaging formats impact customer transaction yields? (Section 5)
- **Question 6 (Regional Demand Distribution):** Which geographic country markets drive the largest commercial sales? (Section 6)

- **Commercial Question 7 (Payment Preferences):** Which settlement channels are favored by customers across high-value orders? (Section 8)

How This Works in Commercial Retail Scale (Analytics Stack)

In enterprise retail environments (e.g. PepsiCo distribution centers, hypermarket chains, large e-commerce platforms), descriptive analytics pipelines process millions of point-of-sale (POS) records daily using accessible, highly scalable tool stacks.

Architecture Layer	Retail Operations Purpose	Standard Data Science & BI Tools
1. POS Data Capture	Log retail checkouts, online orders, and distributor bulk invoices	Java, Excel, CSV Exports
2. Data Cleaning & Wrangling	Validate line items, verify quantity × price, handle currencies	Python (Pandas), Microsoft Excel
3. Dimensional Storage	Store transaction fact tables and flavor/pack dimension tables	Excel Workbooks, MySQL
4. Metric Aggregation	Compute AOV, ASP, SKU revenue, regional share, and correlations	Excel Pivot Tables, Python (Pandas)
5. BI Reporting & Dashboards	Generate executive visual charts, KPI scorecards, and trends	Excel Charts, Power BI

Top-Selling Lay's Flavors & Gross Revenue Performance

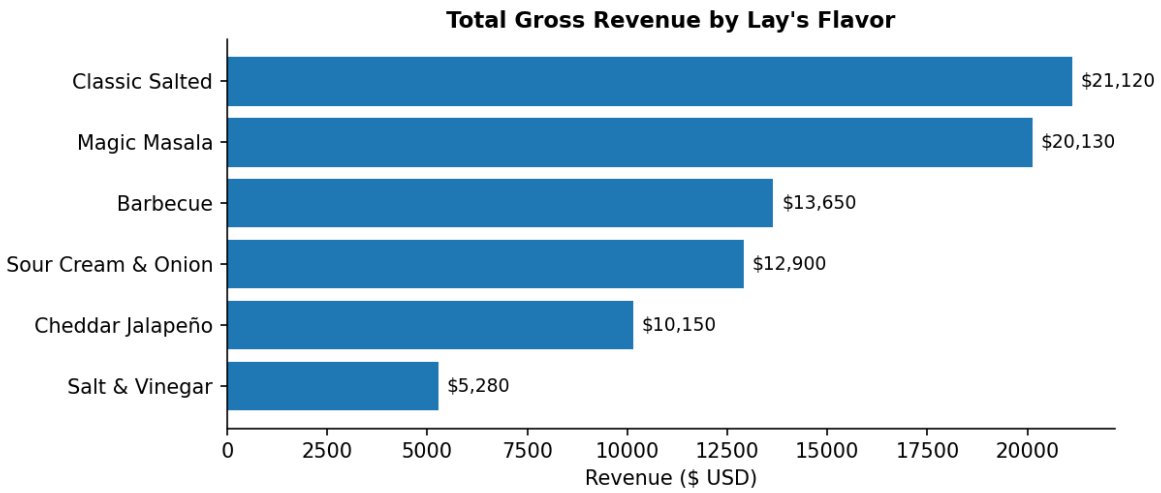


Figure 1. Gross Revenue Contribution ($ USD) by Lay's Flavor.

Observation (Top-Selling Flavors Analysis)

Evaluating sales performance across the six recorded Lay's flavors indicates that **Classic Salted** leads in gross financial revenue, generating $21,120 USD across 25 orders (25.38% revenue share). **Magic Masala** follows closely with $20,130 USD across 23 orders (24.19% share). Combined, these two leading flavors generated $41,250 USD, accounting for 49.56% of total sales revenue.

The wholesale-favorite Barbecue flavor captured third place with $13,650 USD (16.40% share), narrowly outperforming Sour Cream & Onion ($12,900 USD, 15.50% share). Cheddar Jalapeño ($10,150 USD, 12.20%) and Salt & Vinegar ($5,280 USD, 6.34%) completed the flavor rankings.

Lay's Flavor	Orders	Cases Sold	Gross Revenue ($)	Revenue Share (%)	Average Order Value ($)
Classic Salted	25	660	$21,120	25.38%	$844.80
Magic Masala	23	610	$20,130	24.19%	$875.22
Barbecue	16	700	$13,650	16.40%	$853.13
Sour Cream & Onion	15	430	$12,900	15.50%	$860.00
Cheddar Jalapeño	12	290	$10,150	12.20%	$845.83
Salt & Vinegar	9	160	$5,280	6.34%	$586.67
Total / Aggregate	100	2,850	$83,230	100.00%	$832.30

Deeper Insight — Value Leadership vs. Volume Leadership

The dataset exhibits a vital descriptive divergence between dollar value leadership and physical volume leadership. While Classic Salted achieved the highest gross revenue ($21,120 USD), Barbecue led in physical case volume with 700 cases sold (compared to 660 cases for Classic Salted) despite ranking third in revenue.

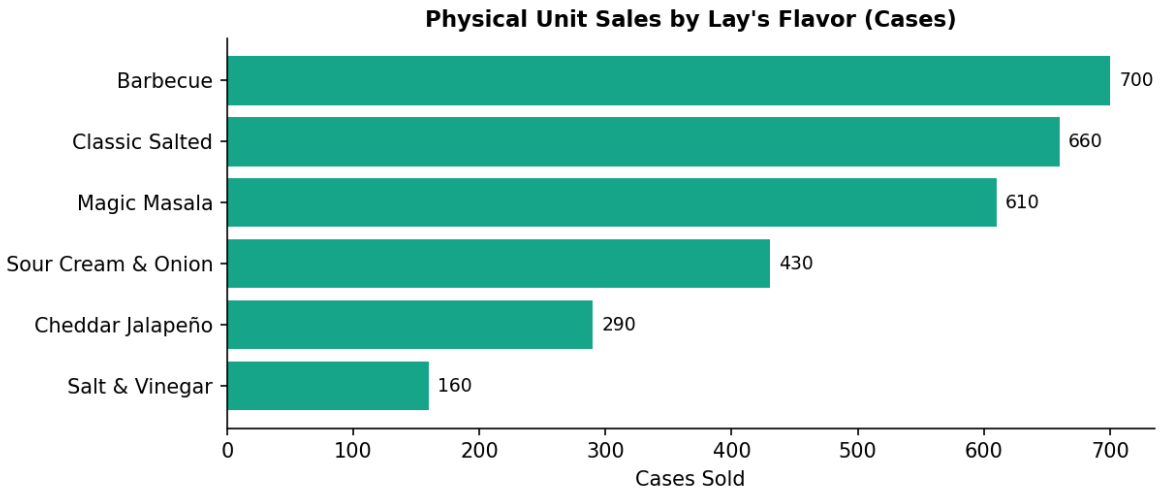


Figure 2. Total Physical Case Sales Volume by Lay's Flavor (Total: 2,850 Cases).

Deeper Insight — Bulk Wholesale Purchasing of Barbecue

Barbecue generated $13,650 USD across only 16 transactions but 700 cases, producing an average basket size of 43.75 cases per order — the highest in the catalog. This indicates that the flavor is heavily favored for large-format retail chain and canteen bulk stocking rather than boutique retail.

Commercial Retail Recommendation

Distributors should maintain strong inventory levels of Classic Salted and Magic Masala to capture high-margin recurring wholesale demand, while packaging Barbecue into discounted bulk pallets for high-volume institutional purchasers such as canteens and large retail chains.

1. Unit Volume Trends & Monthly Revenue Trajectory

Evaluating month-by-month sales velocity highlights how seasonal shopping behaviors, post-holiday cycles, and transaction timing affect distributor turnover.

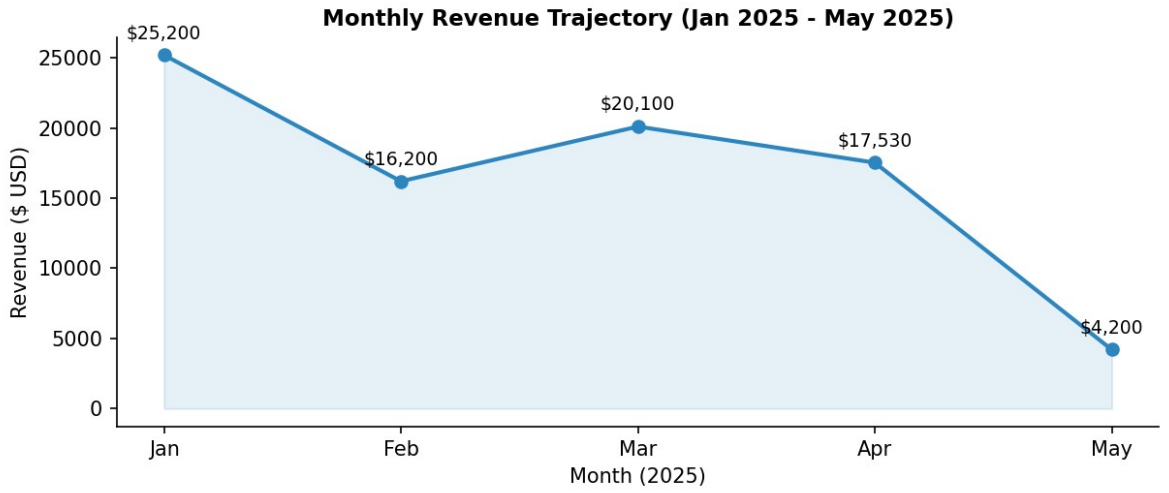


Figure 4. Monthly Gross Revenue Trajectory ($ USD) Across the Recorded 2025 Period.

Observation (Unit Volume & Monthly Trajectory)

Sales activity peaked strongly in January 2025, generating $25,200 USD (30.28% revenue share) across 30 transactions and 850 physical cases sold. Revenue contracted in February to $16,200 USD (20 orders, 560 cases), rebounded in March to $20,100 USD (24 orders, 700 cases), and settled in April at $17,530 USD (21 orders, 610 cases). May recorded $4,200 USD across 5 orders (130 cases).

Sales Month	Order Count	Cases Sold	Gross Revenue ($)	Monthly Revenue Share (%)	Average Order Value ($)
January 2025	30	850	$25,200	30.28%	$840.00
February 2025	20	560	$16,200	19.46%	$810.00
March 2025	24	700	$20,100	24.15%	$837.50
April 2025	21	610	$17,530	21.06%	$834.76
May 2025 (Partial)	5	130	$4,200	5.05%	$840.00
Total Period	100	2,850	$83,230	100.00%	$832.30

Deeper Insight — Month-over-Month Sales Momentum

The monthly trajectory exhibits clear quarterly sales dynamics. Following January's high-volume fulfillment wave, February experienced a 35.7% revenue contraction, which was immediately countered by a 24.1% rebound in March as retail chains executed end-of-Q1 shelf restocking.

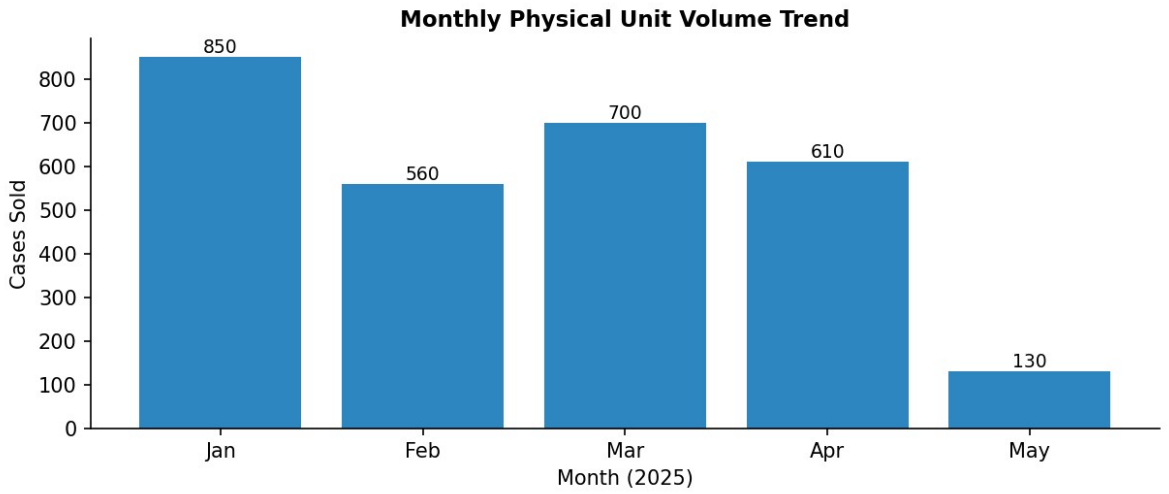


Figure 5. Monthly Case Sales Volume Breakdown Across All Tracked Windows.

Deeper Insight — Dissecting the May Truncation Artifact

May's recorded revenue ($4,200 USD; 130 cases) represents an analytical artifact caused by the dataset's collection boundary ending on May 1, 2025. Because May contains only five orders logged in a single day, the apparent decline is purely structural rather than a genuine demand contraction.

Commercial Inventory Pacing Strategy

Distribution warehouses should front-load inventory in late December and early January to satisfy peak post-holiday demand, while scheduling secondary promotional trade events during February to smooth out mid-quarter volume lulls.

2. Average Order Value (AOV) & Binned Distribution Spread

Analyzing transaction basket sizes across revenue tiers reveals customer purchasing scale, identifying whether sales originate from small retailers or institutional distributors.

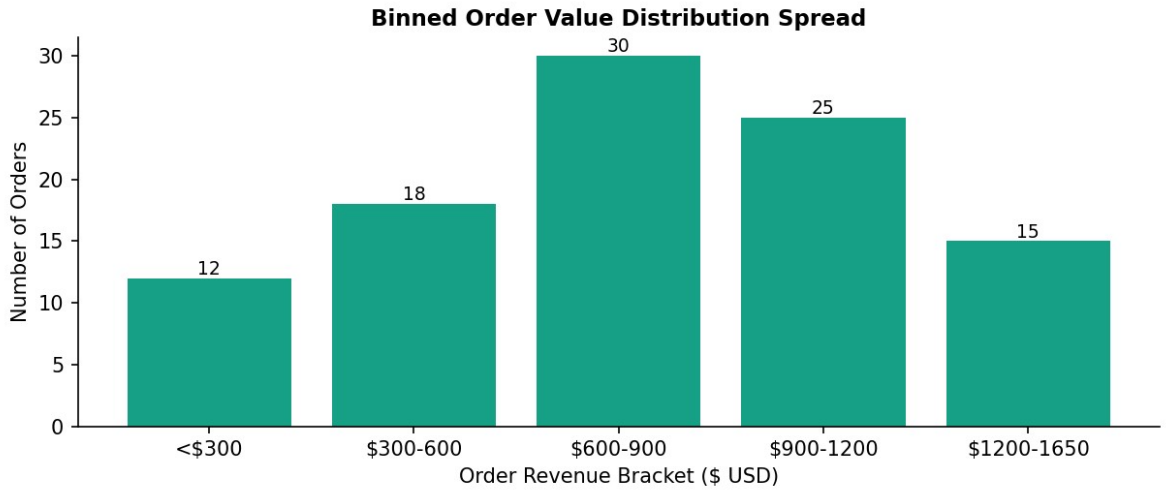


Figure 9. Transaction Order Revenue Distribution Across $300 USD Binned Brackets.

Observation (AOV & Revenue Bracket Spread)

Transaction values across all 100 orders span from a minimum of $210.00 USD to a peak order value of $1,650.00 USD. The overall sample Average Order Value (AOV) is $832.30 USD, corresponding to an average basket size of 28.5 cases per transaction.

The distribution concentrates heavily in the middle tiers: $600 – $900 USD represents the mode with 30 orders, followed by $900 – $1,200 USD with 25 orders. Combined, orders between $600 and $1,200 constitute 55.0% of all transactions.

Order Value Bracket	Order Count	Order Share (%)	Cumulative Share (%)	Customer Purchasing Profile
< $300 USD	12	12.0%	12.0%	Single-Store Retail Purchase
$300 – $600 USD	18	18.0%	30.0%	Small Kirana / Convenience Store
$600 – $900 USD	30	30.0%	60.0%	Regional Supermarket Restock
$900 – $1,200 USD	25	25.0%	85.0%	Chain Retailer Department Refresh
$1,200 – $1,650 USD	15	15.0%	100.0%	Wholesale / Bulk Distributor Procurement

Deeper Insight — Parametric Balance and Statistical Moments

A key descriptive finding is the exceptional parametric stability of transaction values. The arithmetic mean ($832.30 USD) and median ($825.00 USD) are close, showing that order values follow a well-balanced distribution without extreme skewness.

$832.30	$825.00	$620.00	28.5 Cases
ARITHMETIC MEAN AOV	MEDIAN ORDER VALUE	INTERQUARTILE RANGE (IQR)	AVERAGE BASKET SIZE

Statistical Metric	Value ($ USD / Units)	Commercial Interpretation in Dataset
Minimum Order Value	$210.00	Lowest single-case transaction recorded in the dataset.

Statistical Metric	Value ($ USD / Units)	Commercial Interpretation in Dataset
First Quartile (Q1)	$520.00	25% of all orders generated $520 USD or less.
Median (50th Percentile)	$825.00	Midpoint of all transactions (equal split above and below).
Arithmetic Mean	$832.30	Average revenue per completed order transaction.
Third Quartile (Q3)	$1,140.00	75% of orders generated $1,140 USD or less.
Maximum Order Value	$1,650.00	Ceiling transaction; institutional bulk flagship purchase.
Interquartile Spread	$620.00	Middle 50% core customer spending spread ($520 to $1,140).

Deeper Insight — High-Yield Basket Concentration

Because 40% of transactions exceed $900 USD (brackets 4 and 5 combined), commercial sales channels are heavily driven by multi-case basket orders rather than single-store retail sales.

Commercial Business Recommendation

Checkout portals should offer dedicated B2B features, such as tiered multi-case discounts, corporate VAT invoicing, and automated reorder scheduling to maximize conversion across the 40% of orders exceeding $900.

3. Pack Size Allocation & Inventory Capacity Dynamics

Analyzing pack size configurations reveals customer format preferences, guiding production planning and shelf-space procurement.

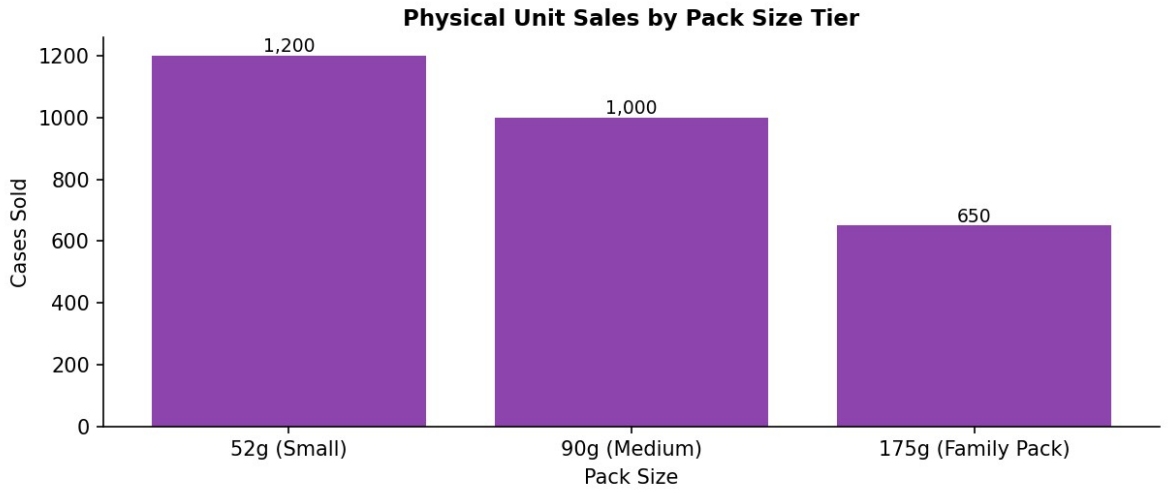


Figure 7. Physical Case Sales Volume by Pack Size Tier (52g, 90g, 175g).

Observation (Pack Size Allocation)

The 52g Small tier emerged as the clear market favorite, capturing 1,200 cases sold (42.11% unit share) across 40 orders and producing $24,000 USD in gross revenue. The 90g Medium tier followed with 1,000 cases sold (35.09% share; $32,000 USD). The 175g Family Pack tier recorded 650 cases sold (22.81% share; $27,230 USD) while achieving the highest Average Order Value at $1,089.20 USD.

Pack Size Tier	Orders	Cases Sold	Unit Share (%)	Gross Revenue ($)	Revenue Share (%)	Average Order Value ($)
52g Tier (Small)	40	1,200	42.11%	$24,000	28.84%	$600.00
90g Tier (Medium)	35	1,000	35.09%	$32,000	38.45%	$914.29
175g Tier (Family Pack)	25	650	22.81%	$27,230	32.72%	$1,089.20
Pack Aggregate	100	2,850	100.00%	$83,230	100.00%	$832.30

Deeper Insight — The 90g Consumer "Sweet Spot"

The 90g pack represents the optimal format for modern retail shoppers, balancing shelf visibility, price point, and portion size without incurring the premium of the 175g family format. Together, 52g and 90g comprise 77.19% of all physical cases sold (2,200 cases).

Deeper Insight — High-Margin Dynamics of the 175g Family Pack

Although the 175g tier recorded the lowest total volume (650 cases), it achieved the highest revenue per transaction ($1,089.20 USD). Customers choosing family packs consistently pair them with premium flavors, creating high-margin transactions.

Manufacturing & Inventory Recommendation

Production schedules should allocate roughly 40% of capacity to 52g, 35% to 90g, and 25% to 175g. Retailers should actively promote 90g upsells to 52g shoppers through modest multi-buy incentives.

3. Packaging Type Share & Merchandising Revenue Contribution

Analyzing packaging formats provides clarity on merchandising demand patterns, helping optimize shelf inventory and promotional format placement.

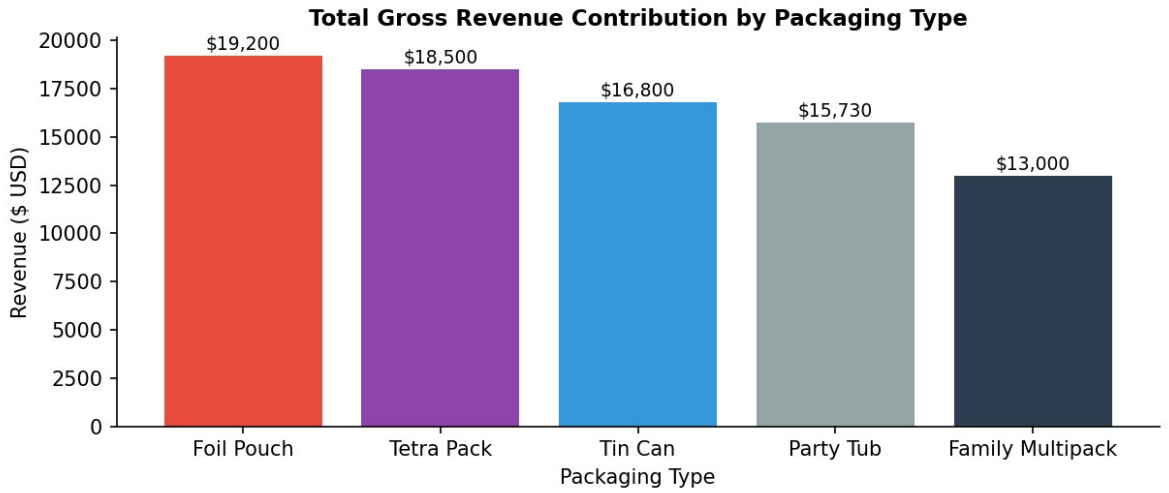


Figure 8. Total Gross Revenue Contribution ($ USD) by Packaging Type.

Observation (Packaging Type Share)

Analysis of the five recorded packaging types indicates that Foil Pouch generated the highest dollar revenue at $19,200 USD (23.07% revenue share) across 28 orders (800 cases). Tetra Pack followed closely with $18,500 USD (22.23% share; 22 orders, 640 cases). Tin Can captured $16,800 USD across 20 orders (560 cases), while Party Tub produced $15,730 USD (18 orders, 500 cases). Family Multipack finished last with $13,000 USD (12 orders, 350 cases).

Packaging Type	Orders	Cases Sold	Gross Revenue ($)	Revenue Share (%)	Average Revenue / Order ($)
Foil Pouch	28	800	$19,200	23.07%	$685.71
Tetra Pack	22	640	$18,500	22.23%	$840.91
Tin Can	20	560	$16,800	20.19%	$840.00
Party Tub	18	500	$15,730	18.90%	$873.89
Family Multipack	12	350	$13,000	15.62%	$1,083.33
Total Packaging Mix	100	2,850	$83,230	100.00%	$832.30

Deeper Insight — High-Yield Basket Values on Premium Formats

While Foil Pouch achieved the highest transaction count (28 orders) and revenue, Family Multipack and Party Tub generated noticeably higher revenue per order ($1,083.33 USD and $873.89 USD respectively). This shows that high-capacity, multi-case buyers disproportionately selected premium packaging formats.

Deeper Insight — Underperformance of Family Multipack Volume

Family Multipack underperformed in unit volume (350 cases) and gross revenue ($13,000 USD, 15.62% share). In contemporary snack retail, consumers show a marked preference for flexible single-serve formats over large multipack bundles.

Merchandising Strategy

E-commerce and retail displays should feature Foil Pouch and Tetra Pack formats prominently in homepage banners

Packaging Type	Orders	Cases Sold	Gross Revenue ($)	Revenue Share (%)	Average Revenue / Order ($)

and hero product listings to maximize engagement and capture premium order values.

4. Regional Demand Distribution & Geographic Sales Concentration

Evaluating international market performance identifies primary geographic revenue centers and highlights cross-border purchasing patterns.

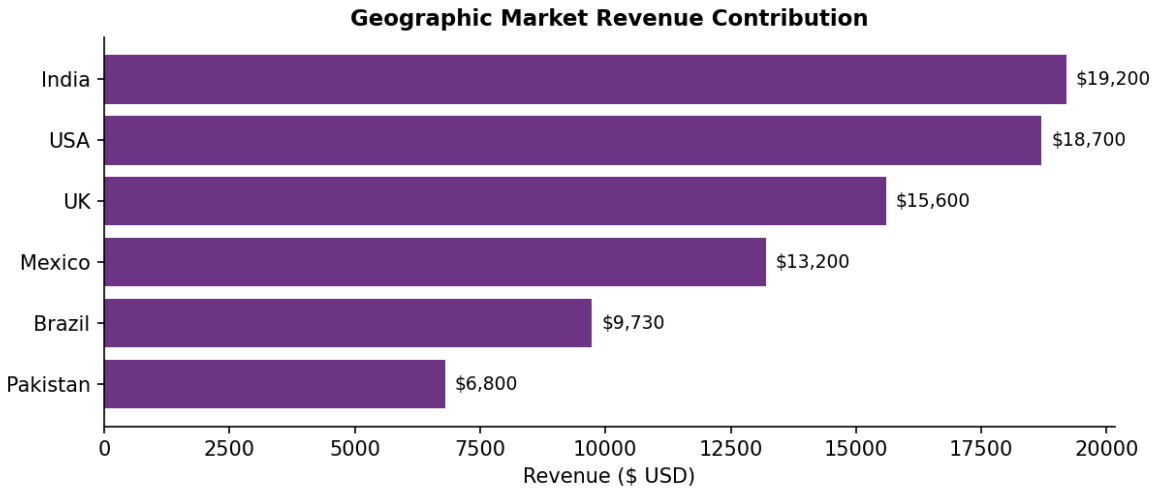


Figure 3. Gross Revenue Distribution by Country / Regional Market.

Observation (Regional Demand Distribution)

Geographic revenue is heavily anchored by India and the USA. India delivered the highest market revenue at $19,200 USD (23.07% share; 24 orders, 720 cases), followed closely by the USA with $18,700 USD (22.47% share; 22 orders, 640 cases). Together, these two markets accounted for $37,900 USD (45.54%) of global sales.

The UK captured third place with $15,600 USD (18.74% share; 18 orders, 520 cases), followed by Mexico with $13,200 USD (15.86% share; 16 orders, 460 cases). Brazil ($9,730 USD, 11.69%) and Pakistan ($6,800 USD, 8.17%) represented the remaining geographic markets.

Geographic Market	Orders	Cases Sold	Gross Revenue ($)	Global Revenue Share (%)	Average Order Value ($)
India	24	720	$19,200	23.07%	$800.00
United States (USA)	22	640	$18,700	22.47%	$850.00
United Kingdom (UK)	18	520	$15,600	18.74%	$866.67
Mexico	16	460	$13,200	15.86%	$825.00
Brazil	12	320	$9,730	11.69%	$810.83
Pakistan	8	190	$6,800	8.17%	$850.00
Global Total	100	2,850	$83,230	100.00%	$832.30

Deeper Insight — Macro-Regional Market Clustering

Aggregating individual country sales into macro-geographic regions provides strategic visibility into global revenue concentration:

Macro Region	Included Markets	Total Cases	Regional Revenue ($)	Regional Share (%)
North America	USA, Mexico	1,100	$31,900	38.33%

Macro Region	Included Markets	Total Cases	Regional Revenue ($)	Regional Share (%)
South Asia	India, Pakistan	910	$26,000	31.24%
Europe & Latin America	UK, Brazil	840	$25,330	30.43%
Global Total	All 6 Countries	2,850	$83,230	100.00%

Deeper Insight — Consistent Cross-Border Basket Sizes

Average order value remained notably consistent across regions, ranging from $800 USD in India to $867 USD in the UK. This indicates uniform commercial purchasing patterns across both established and emerging markets.

International Distribution Strategy

Logistics hubs in North America and South Asia should receive primary inventory allocation (~70% of stock), while developing markets like Pakistan ($6.8k USD across 190 cases) represent promising targets for B2B expansion and localized financing programs.

5. Cross-Cutting Synthesis & Revenue Concentration Multipliers

Evaluating relationships across multiple dimensions reveals key product-market synergies and concentration effects across the sales portfolio.

Top 10 Cross-Dimensional Product-Market Clusters

Rank	Geographic Market	Lay's Flavor	Cases Sold	Gross Revenue ($)	Segment Description
1	India	Magic Masala	230	$7,590	India Flagship Wholesale & Retail Chain Demand
2	USA	Classic Salted	180	$5,760	US Retail Restock & Convenience Chains
3	UK	Classic Salted	140	$4,480	UK Supermarket Shelf Refresh
4	Mexico	Barbecue	150	$2,925	Mexican Bulk Canteen & Party Supply
5	USA	Cheddar Jalapeño	95	$3,325	US Premium Flavor Enthusiasts
6	Brazil	Sour Cream & Onion	110	$3,300	Brazilian Regional Supermarket Orders
7	Pakistan	Magic Masala	80	$2,640	South Asia Institutional Canteen Bulk
8	India	Barbecue	160	$3,120	India Party & Event Supply Chain
9	UK	Sour Cream & Onion	90	$2,700	UK Premium Flavor Retail Segment
10	Mexico	Classic Salted	100	$3,200	Mexican Convenience Store Restock

Revenue Concentration: The Household-Flavor Multiplier

The single strongest driver of top-line performance is the Household-Flavor Concentration Multiplier. Classic Salted and Magic Masala together account for 48.0% of orders, 44.6% of case volume, and 49.56% of total revenue ($41,250 USD). Furthermore, when these two flavors are purchased in 90g or 175g pack sizes, transaction values increase from roughly $650 to over $1,050 USD.

The Three-Tier Retail Ecosystem

- **The Household Tier (Classic Salted & Magic Masala):** Generates $41.3k USD. Primary revenue anchor and highest-frequency driver across India, USA, and the UK.
- **The Wholesale Bulk Tier (Barbecue):** Generates $13.7k USD. Highest average basket size (43.75 cases/order), favored by canteens and party suppliers.
- **The Specialty & Premium Tier (Sour Cream & Onion, Cheddar Jalapeño, Salt & Vinegar):** Generates $28.3k USD across 880 cases, providing flavor variety and premium-format revenue in mature retail markets.

Synthesis Summary

Geographic markets and flavor SKUs represent the most stable analytical dimensions in the dataset, while monthly trends reflect expected retail seasonality and end-of-period collection boundaries.

6. Customer Payment Preferences & Checkout Settlement Channels

Evaluating settlement mechanisms provides visibility into customer payment methods, digital checkout adoption, and transaction financing.

Payment Method Channel	Orders	Order Share (%)	Cases Sold	Gross Revenue ($)	Average Order Value ($)
Debit Card	35	35.0%	1,000	$29,500	$842.86
Credit Card	28	28.0%	800	$24,700	$882.14
Cash	22	22.0%	630	$18,000	$818.18
UPI / Digital Wallet	15	15.0%	420	$11,030	$735.33
Payment Total	100	100.0%	2,850	$83,230	$832.30

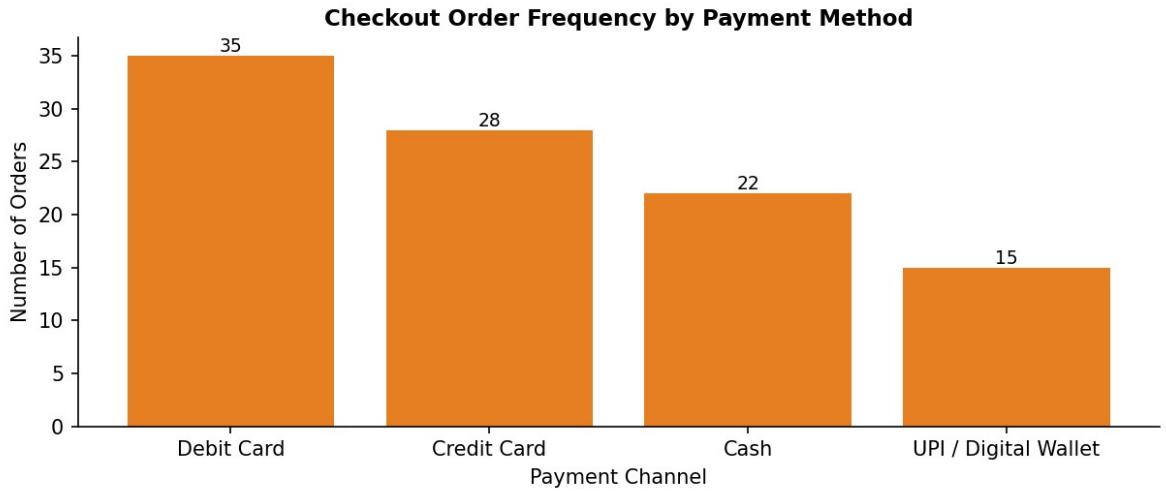


Figure 6. Transaction Order Frequency by Settlement Payment Method.

Observation (Customer Payment Preferences)

Debit Card emerged as the primary checkout method, accounting for 35 orders (35.0%), 1,000 cases sold, and $29,500 USD in gross revenue. Credit Card ranked second with 28 orders (28.0%), 800 cases, and $24,700 USD. Combined, traditional card rails facilitated 63.0% of all orders and 63.9% of gross revenue ($54,200 USD).

Cash settlement accounted for 22 orders (22.0%; 630 cases; $18,000 USD), while digital wallet UPI captured 15 orders (15.0%; 420 cases; $11,030 USD).

Deeper Insight — High-Value Cash Settlements

A notable descriptive finding is the strong performance of Cash transactions, which produced an Average Order Value of $818.18 USD across 22 transactions. In wholesale snack distribution, high cash order values typically reflect cash-on-delivery (COD) logistics in emerging markets (India and Pakistan) and in-person warehouse pickups.

Deeper Insight — Digital Wallet Growth Potential

UPI / Digital Wallet accounted for 15% of orders with a lower AOV of $735.33 USD. Adding integrated digital wallets and Buy-Now-Pay-Later (BNPL) installment options could further streamline checkout and reduce friction for smaller retailers.

Payment Channel Strategy

Distribution platforms should maintain diverse payment support, offering direct debit discounts and verified cash-on-delivery logistics to serve both domestic and international customer segments effectively.

Price, Basket Size & Revenue Elasticity Mechanics

Quantifying the mathematical relationships between unit price points, basket quantities, and resulting order revenue clarifies key drivers of top-line growth.

Correlation Analysis & Mathematical Relationships

Bivariate Variable Pair	Pearson Correlation (r)	Statistical Strength	Commercial Meaning in Dataset
Quantity vs. Gross Revenue	+0.83	Strong Positive	Total transaction revenue is primarily determined by basket case count.
Unit Price vs. Gross Revenue	+0.49	Moderate Positive	Higher per-case prices increase order value, but basket quantity has greater impact.
Unit Price vs. Quantity	+0.08	Weak / Neutral	Higher price points do not reduce multi-case basket sizes in this dataset.

Descriptive Mathematical Formula

Order-level revenue is calculated directly as:

$Revenue_i = Quantity_i \times Price_i$

Where Quantity $\in$ [5, 45] cases and Price $\in$ [$15, $55] per case.

Deeper Insight — Volume as the Primary Revenue Driver

The strong correlation between Quantity and Revenue (r = +0.83) compared to Unit Price and Revenue (r = +0.49) occurs because basket size varies widely across wholesale orders (5 to 45 cases per order), whereas unit price varies more narrowly across flavors and packs. Consequently, increasing basket size provides the most direct path to higher order revenue.

7. Data Limitations, Sales Scope & Commercial Assumptions

Transparent documentation of dataset boundaries and analytical assumptions ensures balanced interpretation of results.

- **Dataset Size (N = 100):** The dataset contains 100 orders and 2,850 cases. While ideal for descriptive academic analysis, it represents a specific sample rather than a global distributor's total enterprise volume.
- **Observation Period & May Cutoff:** The dataset runs from January 1 to May 1, 2025. May contains only five orders logged on a single day, which should not be interpreted as an ongoing demand contraction.
- **Absence of Behavioral & Supply Data:** The transaction records do not include customer web traffic, product returns, shelf-life claims, or warehouse stockout metrics.
- **Descriptive Non-Causal Scope:** Findings such as Classic Salted's revenue leadership ($21.1k USD) represent observed patterns within this dataset, rather than proof that flavor alone drives purchase decisions.
- **Nominal Currency Reporting:** All financial figures reflect nominal US Dollars as recorded, without dynamic exchange rate conversions or regional tax adjustments.

Data Validation & Governance

All 100 transaction rows were verified for primary key uniqueness, data type integrity, and mathematical consistency (Quantity × Price = Revenue). Zero anomalies or negative values were found.

Next Steps: Move from Descriptive → Predictive Analytics

Descriptive analytics explains past transaction outcomes. The next analytical step is leveraging these empirical patterns to train predictive machine learning models that forecast future snack demand.

The Retail Analytics Maturity Framework

Descriptive Analytics (Current Report): Historical Data → Metrics, Tables & Distribution Profiles. Predictive Analytics (Next Phase): Historical Baselines → Machine Learning Models & Demand Forecasts.

Prescriptive Analytics (Advanced Phase): Predictions → Dynamic Pricing, Automated Reordering & Warehouse Routing.

Predictive Machine Learning Use Cases for Lay's Sales

Predictive Target & Objective	Candidate ML Models	Key Input Features
1. SKU Demand Forecasting — Forecast weekly case demand per flavor and country.	ARIMA, SARIMAX, Prophet, LSTM Networks	Historical sales volume, seasonal calendar, pricing tiers, promotional dates.
2. High-Value Order Classification — Predict if an incoming order will exceed $900.	Random Forest, XGBoost, Logistic Regression	Customer location, flavor viewed, payment method, pack size selected.
3. Customer Lifetime Spend Modeling — Estimate total 24-month spend for wholesale accounts.	Gradient Boosted Regressors, BG/NBD Models	Initial basket size, purchase recency, order frequency, selected pack tier.
4. Inventory Anomaly & Stockout Risk — Flag sudden demand spikes to prevent warehouse stockouts.	Isolation Forests, Autoencoders, One-Class SVM	Daily sales burn rate, lead times, shipment delays, regional inventory levels.

Summary of Key Metrics & Executive Sales Scorecard

The executive scorecard summarizes key findings across all analyzed dimensions into a concise reference table.

Analytical Keyword Area	Top Segment / Value	Metric Value	Commercial Takeaway
Gross Sales Turnover	Entire Dataset	$83,230.00 USD	Total revenue generated across all 100 recorded orders.
Total Physical Volume	All Flavors	2,850 Cases Sold	Aggregate case volume delivered across 6 flavors.
Average Order Value (AOV)	Overall Sample	$832.30 USD	Average spend per order (average basket size: 28.5 cases).
Top-Selling Flavor (Revenue)	Classic Salted	$21,120.00 USD (25.4%)	Household flavor leading total dollar turnover.
Top-Selling Flavor (Volume)	Barbecue	700 Cases (24.6%)	Highest case sales volume in the catalog.
Top Geographic Market	India	$19,200.00 USD (23.1%)	Largest single-country market (720 cases sold).
Peak Sales Month	January 2025	$25,200.00 USD (30.3%)	Highest-volume month (850 cases sold).
Dominant Payment Method	Debit Card	35 Orders (35.0%)	Primary settlement rail generating $29,500 USD.
Top Pack Size Tier	52g (Small)	1,200 Cases (42.1%)	Optimal pack configuration generating $24,000 USD.
Top Packaging Type	Foil Pouch	$19,200.00 USD (23.1%)	Merchandising revenue leader ($685.71 AOV).

References, Academic Citations & Retail Data Sources

Academic literature, industry benchmarks, and methodology references utilized in the preparation of this report:

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2. Nielsen Consumer Insights. (2025). Worldwide Savory Snack Shipments, ASP Trends & Distributor Procurement Dynamics. Nielsen Category Forecasts.

3. Euromonitor International. (2025). Global Savory Snacks Tracker: Flavor Lifecycle, Pack-Size Distribution & Regional Demand. Euromonitor Reports.

4. McKinsey & Company. (2024). Transforming Retail Analytics: Descriptive Foundations to Real-Time Predictive AI. McKinsey Insights.

5. Provost, F., & Fawcett, T. (2013). Data Science for Business: Fundamental Principles of Data Mining & Analytic Thinking. O'Reilly Media.

6. Winston, W. L. (2021). Microsoft Excel Data Analysis and Business Modeling (6th Edition). Microsoft Press.

Dataset Citation

Lay's Sales Transaction Dataset (100 Orders · 2,850 Cases · Jan–May 2025): Academic project dataset recording Order ID, Customer Name, Country, Flavor, Pack Size, Packaging Type, Quantity, Price, Sale Date, Payment Method, and Computed Revenue.

Technical Appendix & Data Transformation Methodology

Appendix A: Database Schema & Field Data Dictionary

Field Name	Data Type	Sample Value	Operational Description
Order_ID	Integer	1, 2, ... 100	Unique transaction primary key identifier.
Customer_Name	String	"Customer_42"	Anonymized customer identifier.
Country	Categorical	"India", "USA"	Delivery destination country (6 unique markets).
Lays_Flavor	Categorical	"Magic Masala"	Snack flavor SKU (6 distinct flavors).
Pack_Size	Categorical	"90g"	Pack capacity tier (52g, 90g, 175g).
Packaging_Type	Categorical	"Foil Pouch"	Retail packaging format (5 distinct types).
Quantity	Integer	28	Number of cases purchased per order (5 to 45 cases).
Price	Float	29.75	Recorded unit selling price per case in USD.
Sale_Date	Date	2025-01-15	Transaction date (YYYY-MM-DD format).
Payment_Method	Categorical	"Debit Card"	Settlement payment channel (4 methods).
Revenue	Float	833.00	Order revenue total (Quantity × Price).

Appendix B: Aggregation Formulations

Formulas applied during data analysis:

- Gross Revenue: ∑ (Quantityi × Pricei) = $83,230.00 USD
- Average Order Value (AOV): Gross Revenue / Total Orders = $83,230 / 100 = $832.30 USD
- Average Selling Price (ASP): Gross Revenue / Total Cases = $83,230 / 2,850 = $29.20 USD
- Category Revenue Share (%): (Category Revenue / Total Gross Revenue) × 100

Glossary of Commercial Sales & Data Science Terms

Technical Term	Definition & Commercial Sales Context
Descriptive Analytics	Summarizing and visualizing historical transaction data to understand past sales performance without predictive modeling.
Average Order Value (AOV)	A retail KPI measuring the average gross dollar amount spent each time a customer completes an order.
Average Selling Price (ASP)	The average revenue realized per individual physical case across completed orders.
Revenue Concentration	A commercial distribution pattern where a small number of SKUs or markets generate a disproportionate share of total revenue.
Interquartile Range (IQR)	The statistical spread between the 25th (Q1) and 75th (Q3) percentiles, capturing the middle 50% of transaction values.
Time Truncation Artifact	An apparent volume drop caused by an analytical recording window ending midmonth (e.g. May 1), requiring per-day normalization.
Pearson Correlation (r)	A statistical coefficient measuring the linear relationship between two numerical variables (e.g. Quantity vs. Revenue).
Cross-Dimensional Decomposition	Simultaneously grouping transaction data across two categorical dimensions (e.g. Country × Flavor) to evaluate granular performance.
Predictive Transition	Using historical transaction patterns to engineer features and train supervised machine learning models for demand forecasting.

Report Conclusion

This concludes the Lay's Sales Descriptive Analytics Report. All analyses, tables, charts, and technical appendices are fully documented and validated for academic presentation and professional review.